ROCSCIENCE

Case4_Iter3
Slide2 - An Interactive Slope Stability Program
Date Created: 13/03/2026, 8:34:42 am
Software Version: 9.041

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Slide2 Analysis Information

Case4_Iter3

Project Summary

File Name:	Case4_Iter3.slmd
Slide2 Modeler Version:	9.041
Compute Time:	00h:00m:03.118s
Project Title:	Slide2 - An Interactive Slope Stability Program
Date Created:	13/03/2026, 8:34:42 am

General Settings

Units of Measurement:	Metric Units
Time Units:	days
Permeability Units:	meters/second
Data Output:	Maximum
Failure Direction:	Right to Left

Analysis Options

Slices Type:	Vertical
Analysis Methods Used	
	Bishop simplified
	Spencer
Number of slices:	25
Tolerance:	0.0001
Maximum number of iterations:	100
Check malpha < 0.2:	Yes
Create Interslice boundaries at intersections with water tables and piezos:	Yes
Initial trial value of FS:	1
Steffensen Iteration:	Yes

Groundwater Analysis

Groundwater Method:	Water Surfaces
Pore Fluid Unit Weight [kN/m3]:	9.81
Advanced Groundwater Method:	None

Random Numbers

Pseudo-random Seed:	10116
Random Number Generation Method:	Park and Miller v.3

Surface Options

Surface Type:	Circular
Search Method:	Auto Refine Search
Divisions along slope:	30
Circles per division:	15
Number of iterations:	3
Divisions to use in next iteration:	50%
Composite Surfaces:	Disabled
Minimum Elevation:	Not Defined
Minimum Depth:	Not Defined
Minimum Area:	Not Defined
Minimum Weight:	Not Defined

Seismic Loading

Advanced seismic analysis:	No
Staged pseudostatic analysis:	No

Materials

Material 1	
Color	
Strength Type	Mohr-Coulomb
Unit Weight	16 kN/m3
Cohesion	9 kPa
Phi	32 °
Water Surface	None
Ru Value	0

Global Minimums

Method: bishop simplified

FS	1.235220
Center:	20.894, 67.292
Radius:	43.235
Left Slip Surface Endpoint:	30.047, 25.038
Right Slip Surface Endpoint:	57.938, 45.000
Resisting Moment:	54109.1 kN-m
Driving Moment:	43805.2 kN-m
Total Slice Area:	110.962 m2
Surface Horizontal Width:	27.8908 m
Surface Average Height:	3.97844 m

Method: spencer

FS	1.232230
Center:	20.894, 67.292
Radius:	43.235
Left Slip Surface Endpoint:	30.047, 25.038
Right Slip Surface Endpoint:	57.938, 45.000
Resisting Moment:	53978.3 kN-m
Driving Moment:	43805.2 kN-m
Resisting Horizontal Force:	1009.67 kN
Driving Horizontal Force:	819.384 kN
Total Slice Area:	110.962 m2
Surface Horizontal Width:	27.8908 m
Surface Average Height:	3.97844 m

Global Minimum Support Data

No Supports Present

Valid and Invalid Surfaces

Method: bishop simplified

Number of Valid Surfaces:	11848
Number of Invalid Surfaces:	0

Method: spencer

Number of Valid Surfaces:	10071
Number of Invalid Surfaces:	1777

Error Codes

Error Code -108 reported for 279 surfaces
Error Code -111 reported for 1498 surfaces

Error Code Descriptions

The following errors were encountered during the computation:
-108 = Total driving moment or total driving force < 0.1. This is to limit the calculation of extremely high safety factors if the driving force is very small (0.1 is an arbitrary number).
-111 = Safety factor equation did not converge

Slice Data

Global Minimum Query (bishop simplified) - Safety Factor: 1.23522

Slice Number	Width [m]	Weight [kN]	Angle of Slice Base [deg]	Base Material	Base Cohesion [kPa]	Base Friction Angle [deg]	Shear Stress [kPa]	Shear Strength [kPa]	Base Normal Stress [kPa]	Pore Pressure [kPa]	Effective Normal Stress [kPa]	Base Vertical Stress [kPa]	Effective Vertical Stress [kPa]
1	1.11563	5.67027	12.9816	Material 1	9	32	8.8278	10.9043	3.0475	0	3.0475	5.08257	5.08257
2	1.11563	16.7304	14.5038	Material 1	9	32	13.1514	16.2449	11.5942	0	11.5942	14.9964	14.9964
3	1.11563	27.224	16.0366	Material 1	9	32	17.1386	21.17	19.476	0	19.476	24.4022	24.4022
4	1.11563	37.1383	17.5813	Material 1	9	32	20.7933	25.6843	26.7005	0	26.7005	33.289	33.289
5	1.11563	46.4591	19.1393	Material 1	9	32	24.1184	29.7915	33.2735	0	33.2735	41.6437	41.6437
6	1.11563	55.1699	20.7121	Material 1	9	32	27.116	33.4942	39.1989	0	39.1989	49.4517	49.4517
7	1.11563	63.2524	22.3015	Material 1	9	32	29.787	36.7935	44.4789	0	44.4789	56.6964	56.6964
8	1.11563	70.6855	23.9092	Material 1	9	32	32.1318	39.6899	49.1141	0	49.1141	63.3591	63.3591
9	1.11563	77.4453	25.5371	Material 1	9	32	34.1495	42.1822	53.1027	0	53.1027	69.4183	69.4183
10	1.11563	83.505	27.1875	Material 1	9	32	35.8385	44.2684	56.4413	0	56.4413	74.8499	74.8499
11	1.11563	88.8337	28.8627	Material 1	9	32	37.1959	45.9451	59.1245	0	59.1245	79.6262	79.6262
12	1.11563	93.3964	30.5654	Material 1	9	32	38.2181	47.2077	61.1451	0	61.1451	83.7161	83.7161
13	1.11563	97.153	32.2985	Material 1	9	32	38.9	48.05	62.4931	0	62.4931	87.0833	87.0833
14	1.11563	100.057	34.0655	Material 1	9	32	39.2356	48.4646	63.1565	0	63.1565	89.6866	89.6866
15	1.11563	102.056	35.8702	Material 1	9	32	39.2174	48.4421	63.1206	0	63.1206	91.4782	91.4782
16	1.11563	103.087	37.7171	Material 1	9	32	38.8365	47.9716	62.3677	0	62.3677	92.4024	92.4024
17	1.11563	103.078	39.6112	Material 1	9	32	38.0823	47.04	60.8768	0	60.8768	92.3937	92.3937
18	1.11563	101.941	41.5588	Material 1	9	32	36.9424	45.632	58.6235	0	58.6235	91.375	91.375
19	1.11563	99.5736	43.5671	Material 1	9	32	35.4022	43.7295	55.5788	0	55.5788	89.253	89.253
20	1.11563	95.8501	45.6448	Material 1	9	32	33.4447	41.3116	51.7093	0	51.7093	85.9154	85.9154
21	1.11563	90.6154	47.8029	Material 1	9	32	31.0503	38.3539	46.9761	0	46.9761	81.2233	81.2233
22	1.11563	83.6749	50.055	Material 1	9	32	28.196	34.8283	41.3339	0	41.3339	75.0022	75.0022
23	1.11563	71.5804	52.4186	Material 1	9	32	23.9806	29.6213	33.001	0	33.001	64.1613	64.1613
24	1.11563	45.535	54.9171	Material 1	9	32	16.2382	20.0578	17.6961	0	17.6961	40.8154	40.8154
25	1.11563	15.6792	57.5824	Material 1	9	32	8.01283	9.89761	1.43648	0	1.43648	14.0541	14.0541

Global Minimum Query (spencer) - Safety Factor: 1.23223

Slice Number	Width [m]	Weight [kN]	Angle of Slice Base [deg]	Base Material	Base Cohesion [kPa]	Base Friction Angle [deg]	Shear Stress [kPa]	Shear Strength [kPa]	Base Normal Stress [kPa]	Pore Pressure [kPa]	Effective Normal Stress [kPa]	Base Vertical Stress [kPa]	Effective Vertical Stress [kPa]
1	1.11563	5.67027	12.9816	Material 1	9	32	11.6715	14.382	8.61296	0	8.61296	11.3036	11.3036
2	1.11563	16.7304	14.5038	Material 1	9	32	16.5935	20.447	18.3191	0	18.3191	22.6117	22.6117
3	1.11563	27.224	16.0366	Material 1	9	32	20.9302	25.7908	26.8709	0	26.8709	32.887	32.887
4	1.11563	37.1383	17.5813	Material 1	9	32	24.7214	30.4624	34.3471	0	34.3471	42.1802	42.1802
5	1.11563	46.4591	19.1393	Material 1	9	32	28.0019	34.5048	40.8163	0	40.8163	50.5343	50.5343
6	1.11563	55.1699	20.7121	Material 1	9	32	30.8024	37.9557	46.3388	0	46.3388	57.9855	57.9855
7	1.11563	63.2524	22.3015	Material 1	9	32	33.1497	40.8481	50.9676	0	50.9676	64.5643	64.5643
8	1.11563	70.6855	23.9092	Material 1	9	32	35.0673	43.211	54.7491	0	54.7491	70.2955	70.2955
9	1.11563	77.4453	25.5371	Material 1	9	32	36.576	45.07	57.7241	0	57.7241	75.1991	75.1991
10	1.11563	83.505	27.1875	Material 1	9	32	37.6939	46.4476	59.9286	0	59.9286	79.2903	79.2903
11	1.11563	88.8337	28.8627	Material 1	9	32	38.4371	47.3633	61.3942	0	61.3942	82.5799	82.5799
12	1.11563	93.3964	30.5654	Material 1	9	32	38.8195	47.8346	62.1484	0	62.1484	85.0746	85.0746
13	1.11563	97.153	32.2985	Material 1	9	32	38.8536	47.8766	62.2156	0	62.2156	86.7765	86.7765
14	1.11563	100.057	34.0655	Material 1	9	32	38.5502	47.5027	61.6172	0	61.6172	87.6838	87.6838
15	1.11563	102.056	35.8702	Material 1	9	32	37.9186	46.7244	60.3716	0	60.3716	87.7901	87.7901
16	1.11563	103.087	37.7171	Material 1	9	32	36.967	45.5519	58.4953	0	58.4953	87.0843	87.0843
17	1.11563	103.078	39.6112	Material 1	9	32	35.7029	43.9942	56.0025	0	56.0025	85.5502	85.5502
18	1.11563	101.941	41.5588	Material 1	9	32	34.1325	42.0591	52.9057	0	52.9057	83.166	83.166
19	1.11563	99.5736	43.5671	Material 1	9	32	32.2615	39.7536	49.2161	0	49.2161	79.903	79.903
20	1.11563	95.8501	45.6448	Material 1	9	32	30.0952	37.0842	44.9442	0	44.9442	75.7245	75.7245
21	1.11563	90.6154	47.8029	Material 1	9	32	27.6386	34.0571	40.0997	0	40.0997	70.5839	70.5839
22	1.11563	83.6749	50.055	Material 1	9	32	24.8967	30.6785	34.6928	0	34.6928	64.4215	64.4215
23	1.11563	71.5804	52.4186	Material 1	9	32	21.2032	26.1272	27.4093	0	27.4093	54.9607	54.9607
24	1.11563	45.535	54.9171	Material 1	9	32	15.0145	18.5014	15.2054	0	15.2054	36.5824	36.5824
25	1.11563	15.6792	57.5824	Material 1	9	32	8.7597	10.794	2.87095	0	2.87095	16.6647	16.6647

Interslice Data

Global Minimum Query (bishop simplified) - Safety Factor: 1.23522

Slice Number	X coordinate [m]	Y coordinate - Bottom [m]	Interslice Normal Force [kN]	Interslice Shear Force [kN]	Interslice Force Angle [deg]
1	30.0473	25.0379	0	0	0
2	31.163	25.295	9.06482	0	0
3	32.2786	25.5836	20.3908	0	0
4	33.3942	25.9043	33.2658	0	0
5	34.5099	26.2578	47.0249	0	0
6	35.6255	26.645	61.0494	0	0
7	36.7411	27.0668	74.7655	0	0
8	37.8568	27.5244	87.6438	0	0
9	38.9724	28.019	99.1996	0	0
10	40.088	28.552	108.993	0	0
11	41.2037	29.1251	116.632	0	0
12	42.3193	29.74	121.773	0	0
13	43.4349	30.3989	124.123	0	0
14	44.5506	31.1041	123.449	0	0
15	45.6662	31.8585	119.578	0	0
16	46.7818	32.6652	112.411	0	0
17	47.8975	33.528	101.928	0	0
18	49.0131	34.4513	88.2067	0	0
19	50.1287	35.4403	71.438	0	0
20	51.2444	36.5015	51.9548	0	0
21	52.36	37.6425	30.2648	0	0
22	53.4756	38.873	7.10152	0	0
23	54.5913	40.2052	-16.5052	0	0
24	55.7069	41.6548	-37.5917	0	0
25	56.8225	43.2432	-47.5841	0	0
26	57.9382	45	0	0	0

Global Minimum Query (spencer) - Safety Factor: 1.23223

Slice Number	X coordinate [m]	Y coordinate - Bottom [m]	Interslice Normal Force [kN]	Interslice Shear Force [kN]	Interslice Force Angle [deg]
1	30.0473	25.0379	0	0	0
2	31.163	25.295	10.8054	6.94277	32.7219
3	32.2786	25.5836	24.0299	15.4399	32.7219
4	33.3942	25.9043	38.7625	24.906	32.7219
5	34.5099	26.2578	54.1996	34.8248	32.7219
6	35.6255	26.645	69.6349	44.7424	32.7219
7	36.7411	27.0668	84.4504	54.2618	32.7219
8	37.8568	27.5244	98.1095	63.0381	32.7219
9	38.9724	28.019	110.151	70.7754	32.722
10	40.088	28.552	120.187	77.2236	32.7219
11	41.2037	29.1251	127.896	82.1766	32.7218
12	42.3193	29.74	133.023	85.4712	32.7219
13	43.4349	30.3989	135.382	86.9866	32.7218
14	44.5506	31.1041	134.85	86.6447	32.7218
15	45.6662	31.8585	131.374	84.4114	32.7219
16	46.7818	32.6652	124.973	80.2989	32.722
17	47.8975	33.528	115.744	74.3688	32.7219
18	49.0131	34.4513	103.866	66.7372	32.722
19	50.1287	35.4403	89.6167	57.5813	32.7219
20	51.2444	36.5015	73.38	47.1487	32.7219
21	52.36	37.6425	55.671	35.7702	32.7219
22	53.4756	38.873	37.1615	23.8773	32.7219
23	54.5913	40.2052	18.7197	12.0279	32.7218
24	55.7069	41.6548	2.63963	1.69604	32.7219
25	56.8225	43.2432	-4.76246	-3.06002	32.7219
26	57.9382	45	0	0	0

Discharge Sections

Entity Information

Group 1

Shared Entities

Type	Coordinates (x,y)
External Boundary	95, 20 95, 45 55, 45 30, 25 10, 25 10, 20